

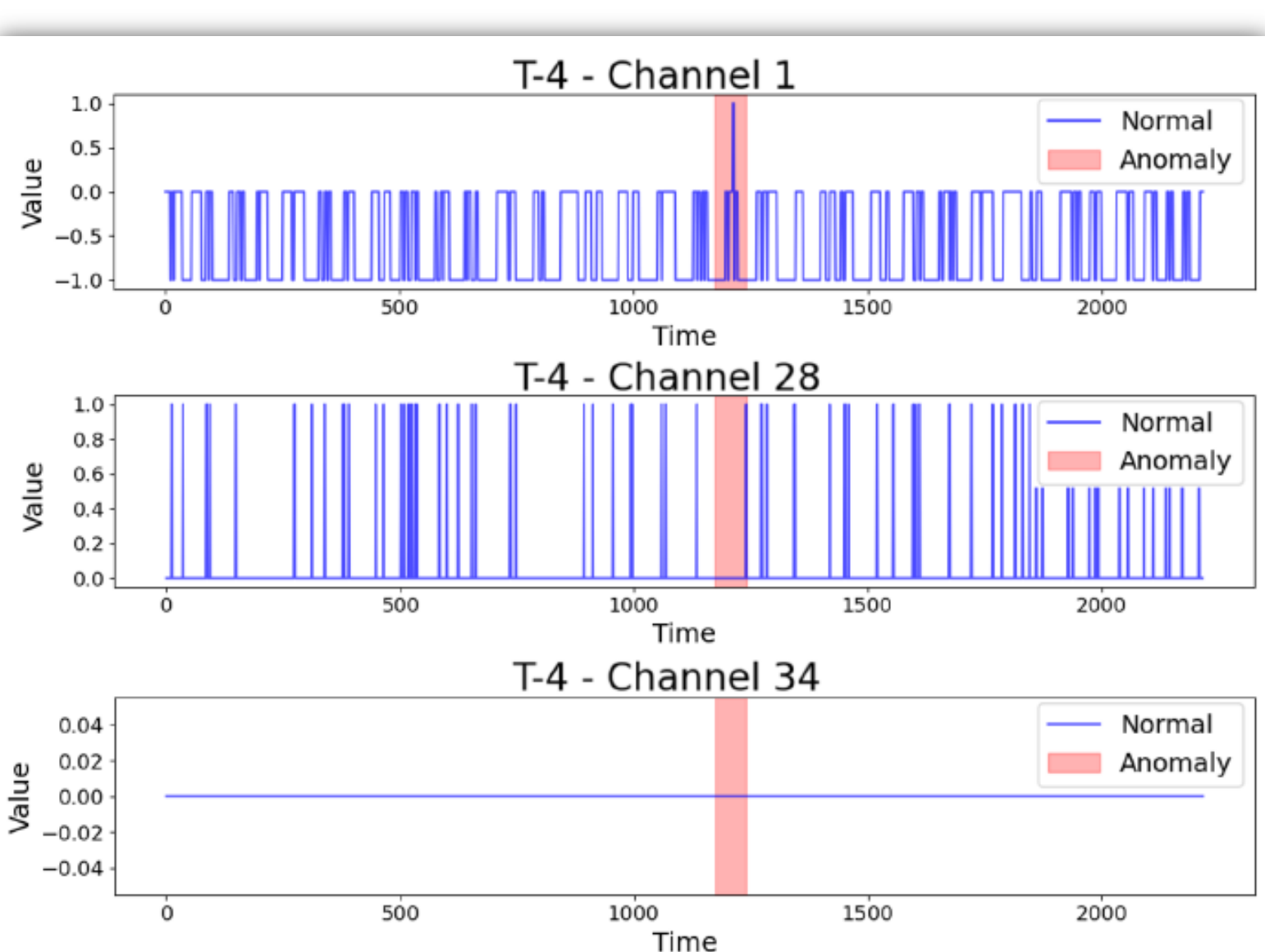
Introduction

Background

- Multivariate time series anomaly detection aims to identify abnormal patterns from multivariate temporal data.
- Contrastive learning is widely used for anomaly detection by pulling similar samples together and pushing dissimilar ones apart.

Motivation

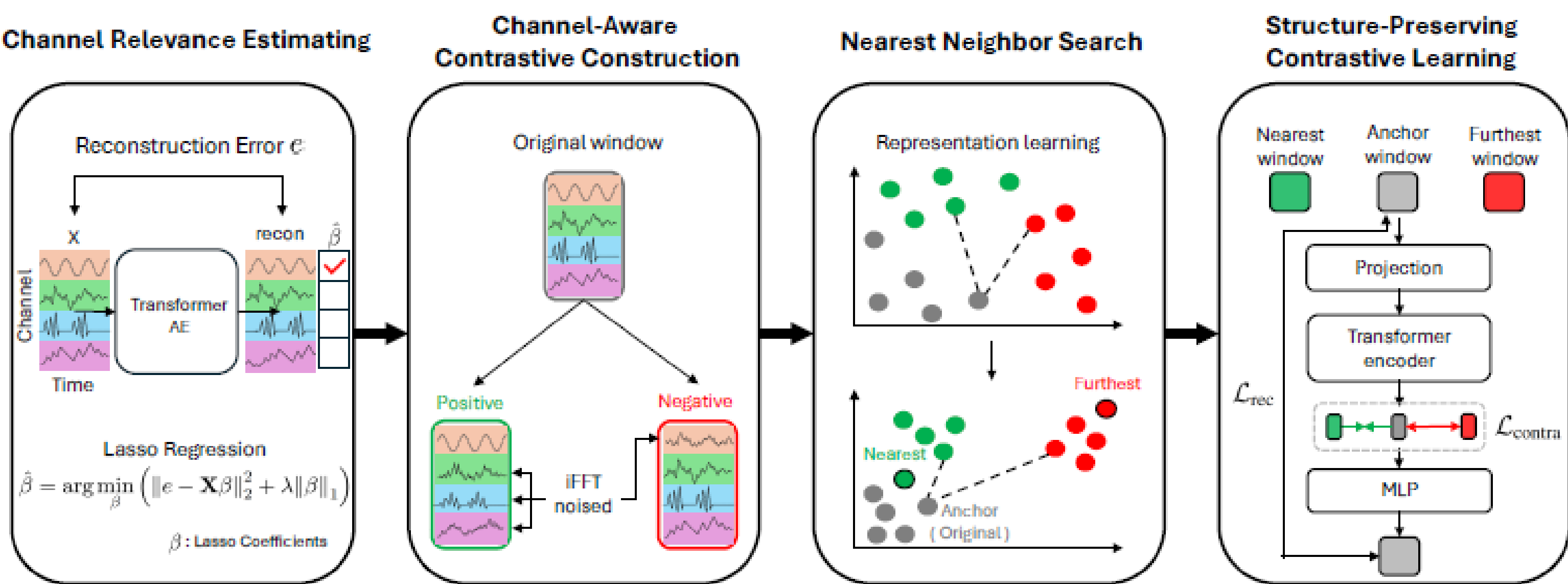
- Not all channels contribute equally to anomaly detection.
- Treating every channel equally may dilute anomaly-relevant information.



An example from the MSL(T-4) dataset showing that each channel exhibits distinct temporal patterns and **different degrees of relevance to anomalies**.

Methods

Overall Framework



Main idea of CALAD

- CALAD **estimates channel relevance** before contrastive learning.
- Channel relevance **guides positive and negative** sample construction.

Details

Channel Relevance Estimation

- Transformer autoencoder computes per-channel reconstruction errors.
- LASSO** identifies anomaly-relevant channels from those errors.

Channel-Aware Contrastive Construction

- Positives** preserve relevant channels while perturbing irrelevant ones.
- Negatives** perturb relevant channels to create anomaly-aware views.

Nearest Neighbor Search

- Positives are pulled closer, while negatives are pushed farther apart.
- The nearest and furthest neighbors are retrieved for each anchor.

Structure-Preserving Contrastive Learning

- Contrastive learning uses the nearest and furthest neighbors.
- An auxiliary reconstruction loss preserves normal temporal structures.

Experiments

Datasets

Dataset	#Train	#Test	Entity	Dim	AR(%)
MSL	58,317	73,729	27	55	10.48
SMAP	135,183	427,617	55	25	12.83
SMD	708,405	708,420	28	38	4.16
SWaT	495,000	449,919	1	51	12.14

- Dim is short for dimension and AR is short for the anomaly rate.

Main Results

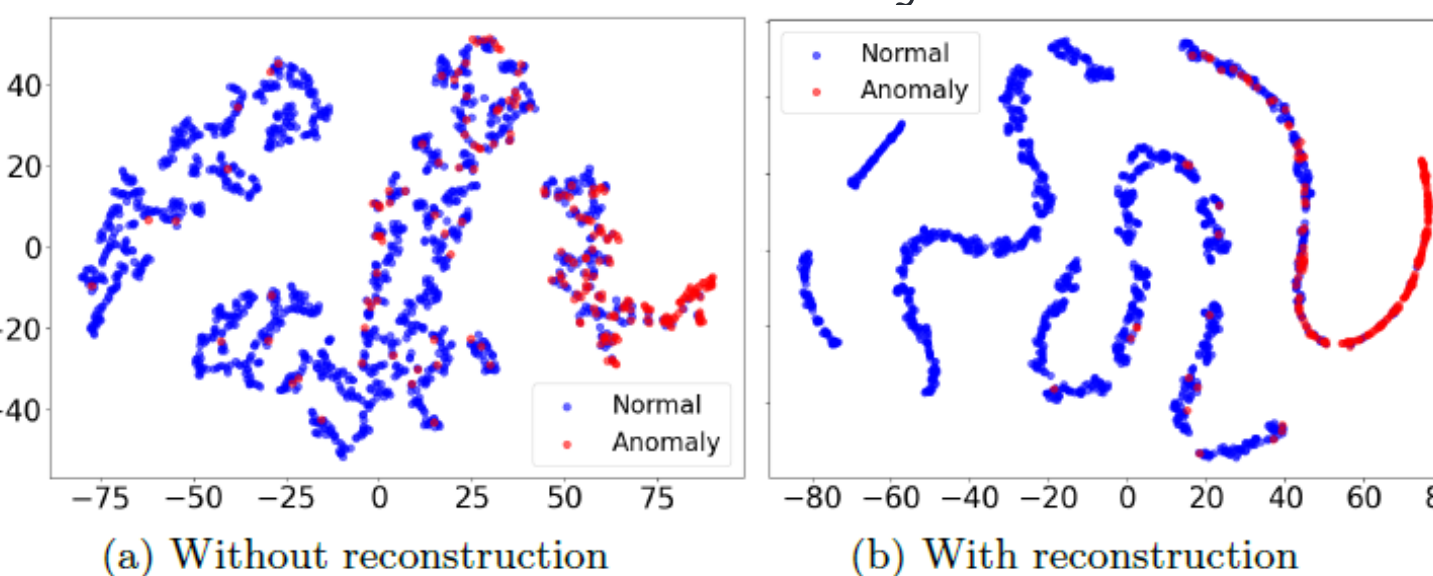
Dataset	MSL		SMAP		SMD		SWaT	
	AU-PR	F1	AU-PR	F1	AU-PR	F1	AU-PR	F1
Random	17.20	29.36	14.00	30.28	8.90	17.31	12.90	21.66
LSTM-VAE	28.50	40.74	25.80	43.70	39.50	29.80	68.50	72.37
OmniAnomaly	22.01	18.45	22.69	18.81	41.66	35.19	21.35	24.25
MSCRED	25.56	21.59	24.49	14.87	41.74	35.37	22.98	13.19
MTAD-GAT	33.50	47.34	33.90	51.86	40.10	34.73	9.50	24.23
THOC	23.90	30.95	19.50	32.73	10.70	16.79	53.70	63.80
AnomTrans	23.60	34.49	26.40	40.74	27.30	30.43	68.10	73.76
UAE	17.41	6.38	15.90	14.75	39.55	23.79	43.44	8.34
TranAD	27.80	42.83	28.70	47.16	41.20	36.09	19.20	31.03
TS2Vec	13.20	29.93	14.80	37.12	11.30	17.28	13.60	26.11
DCdetector	12.90	22.70	12.40	27.53	4.30	8.28	12.60	21.66
TimesNet	28.30	35.78	20.80	40.19	38.50	33.85	8.30	21.66
EdgeConvFormer	32.47	34.04	31.24	17.66	39.67	27.28	70.68	72.75
CARLA	50.10	<u>52.27</u>	44.80	<u>52.92</u>	50.70	<u>51.14</u>	68.10	72.09
CALAD	61.20	68.24	52.30	60.00	51.10	51.53	70.55	<u>72.87</u>

Model Analysis

Entity	Method	P	R	F1
P-15	Augmentation			
	All-channel	99.10	50.00	66.47
	Channel-wise	99.99	64.55	78.45
All-Entities	Noise			
	Anomaly Injection	51.89	74.49	61.12
	iFFT	64.69	72.19	68.24
All-Entities	Backbone			
	ResNet	43.80	74.13	55.07
	Transformer	52.49	73.98	61.41
	Transformer-MLP	64.69	72.19	68.24

$\mathcal{L}_{contra}$	$\mathcal{L}_{recon}$	P	R	F1
—	✓	26.83	79.74	40.15
✓	—	49.61	70.60	58.27
✓	✓	64.69	72.19	68.24

Ablation study

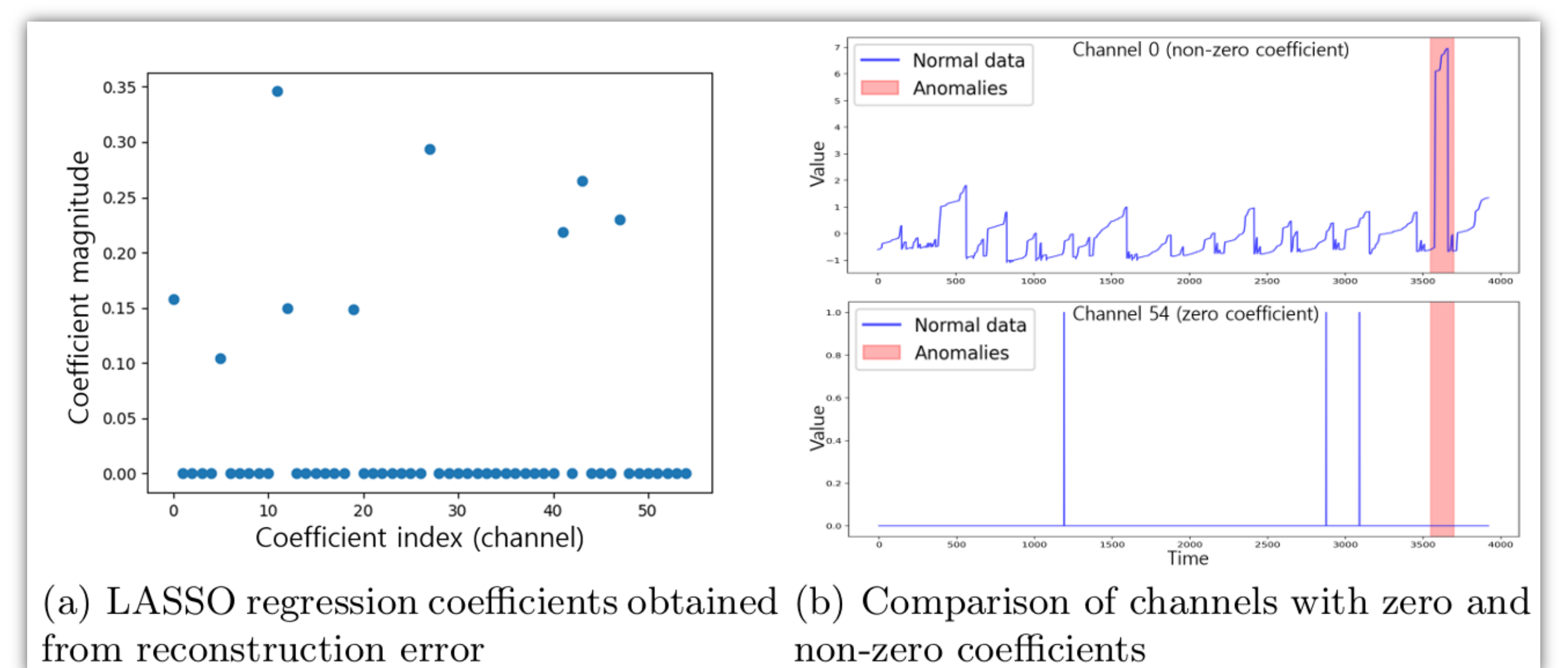


On the MSL

t-SNE embeddings

Channel Relevance Estimation

- Channels with **non-zero coefficients** better reflect anomalies.



Channel relevance estimation on the MSL(F-5)

Conclusion

- ✓ We estimate channel relevance to guide channel-aware contrastive augmentation, unlike channel-agnostic models.
- ✓ Channel-aware construction preserves informative channels during augmentation.



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